

## Introduction

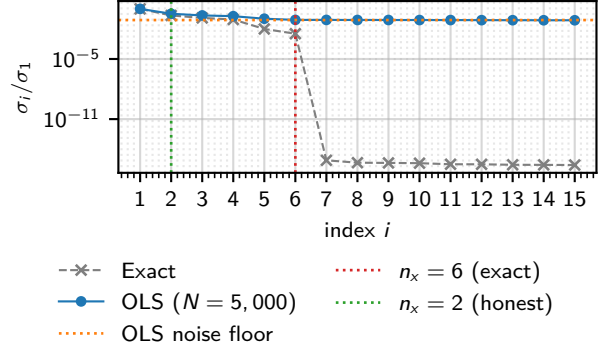
We are provided a brain

## System identification

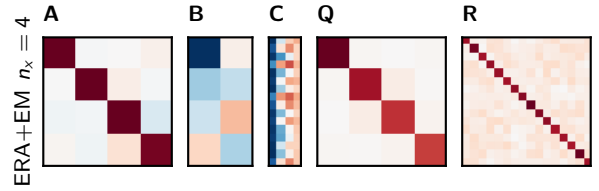
The exact Hankel singular-value spectrum, obtained via same-seed impulse subtraction (a theoretical reference not physically realisable with a real brain), has rank 6 (Figure 1). A physically realisable single-run OLS estimate ( $N = 5,000$  steps) raises a noise floor that obscures modes 3–6 (exact values  $\sigma_3/\sigma_1 = 0.125$  down to  $\sigma_6/\sigma_1 = 3.3 \times 10^{-3}$ ); BIC model selection recovers only  $n_x = 2$ . These modes carry little input–output energy and are near-unobservable from any finite honest recording. However, the downstream decoder operates on frequency-selective latent activity, so a model that misrepresents the transfer function will produce incorrect muscle activations regardless of one-step prediction quality. ERA+EM was therefore fitted at  $n_x \in \{2, 4, 6\}$  to identify the minimum sufficient order.

At  $n_x = 2$  the model reproduces per-neuron frequency selectivity for  $u_1$  but not for  $u_2$  (Figure 5). At  $n_x = 4$  both input channels are recovered;  $n_x = 6$  yields no visible further improvement. ERA+EM at  $n_x = 4$  was therefore selected. The Bode magnitude error (Figure 3), computed diagnostically via same-seed impulse subtraction, confirms this pattern: mean error falls substantially from  $n_x = 2$  to  $n_x = 4$ , with negligible reduction at  $n_x = 6$ .

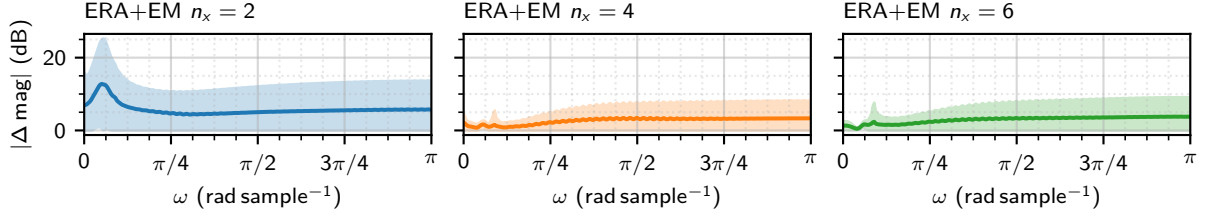
CVA+EM was fitted at  $n_x = 4$  for comparison. Its transfer function reproduces the  $u_1$  selectivity but misrepresents  $u_2$  relative to the honest OLS estimate (rightmost column, Figure 5). One-step-ahead Kalman prediction on  $N = 100$  held-out trials ( $T = 2,000$  steps each) confirmed ERA+EM superiority across all output channels (Figure 4). ERA+EM at  $n_x = 4$  was adopted; the identified matrices are shown in Figure 2.



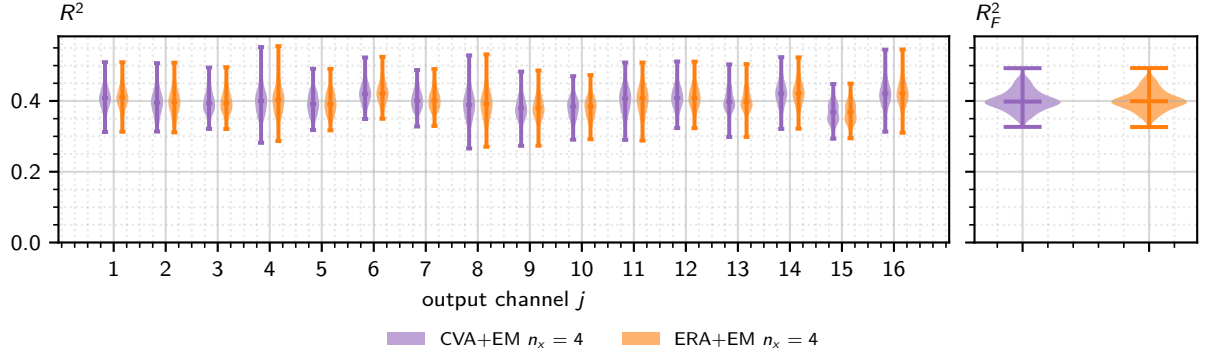
**Figure 1:** Normalised Hankel singular values  $\sigma_i/\sigma_1$  for the block Hankel matrix (40 block rows, 60 block columns,  $n_u = 2$ ,  $n_y = 16$ ). Grey dashed: exact values via same-seed impulse subtraction (not physically realisable; theoretical reference only). Blue solid: OLS estimates from a single honest 5000-step recording. Orange dotted: OLS noise floor (median of OLS values above exact rank). Red dotted:  $n_x = 6$  (exact). Green dotted:  $n_x = 2$  (honest, BIC model selection). Modes 3–6 lie below the honest noise floor, rendering them undetectable from finite recordings. [Source](#)



**Figure 2:** Identified system matrices for ERA+EM ( $n_x = 4$ ). Red indicates positive, blue negative; colour scales are symmetric about zero. [Source](#)



**Figure 3:** Magnitude error of ERA+EM transfer functions for  $n_x \in \{2, 4, 6\}$  (left to right). Each panel shows the absolute magnitude error (dB) between the exact empirical transfer function and the analytic model response  $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ , as mean  $\pm 1$  s.d. across all  $n_u n_y = 32$  output-input channels. The empirical reference is the exact discrete-time frequency response obtained via impulse subtraction (same-seed noise cancellation), DFT zero-padded to 512 points. Panels share a common y-axis. [Source](#)

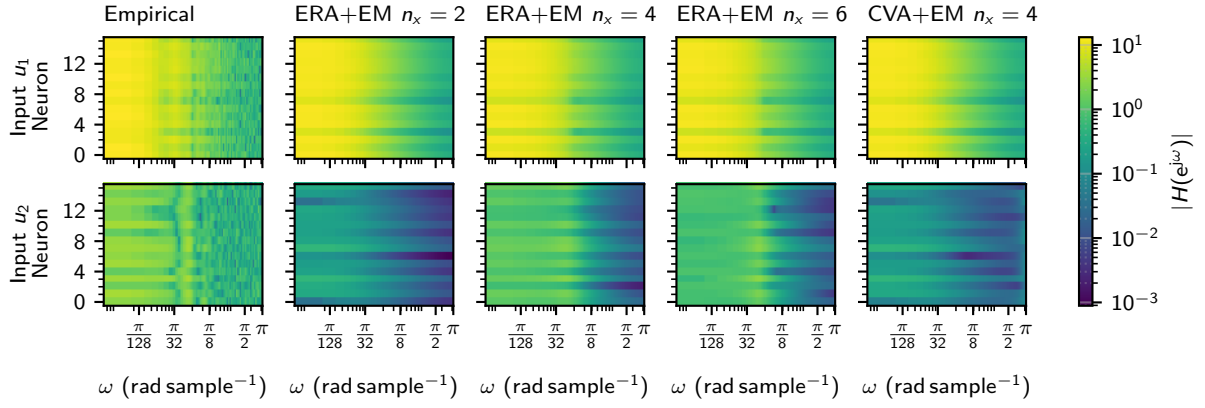


**Figure 4:** Per-channel  $R^2$  (left,  $n_y = 16$ ) and Frobenius  $R_F^2$  (right) of one-step-ahead Kalman predictions for CVA+EM and ERA+EM at  $n_x = 4$ , across  $N = 100$  held-out trials of  $T = 2000$  steps (first 100 steps excluded as Kalman transient). Median: CVA+EM  $n_x = 4$   $R_F^2 = 0.398$ ; ERA+EM  $n_x = 4$   $R_F^2 = 0.399$ . [Source](#)

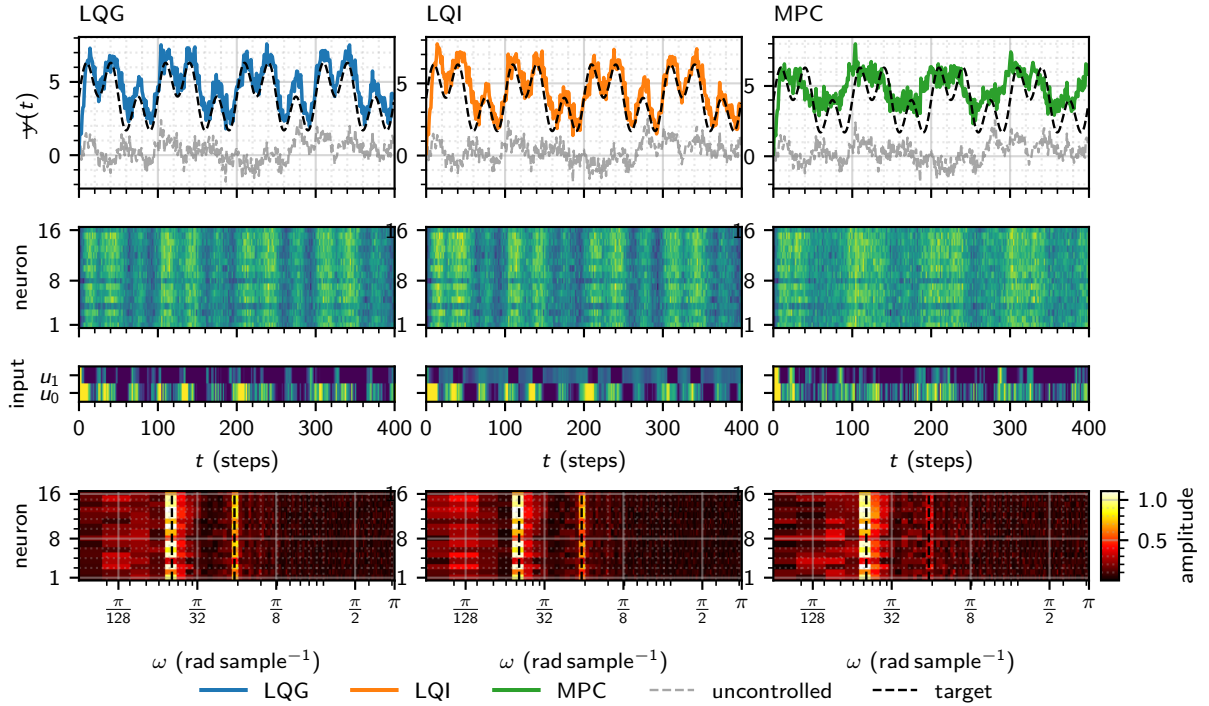
### Closed-loop spectral control

Having selected ERA+EM at  $n_x = 4$  as the working model, we test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population activity. The target is a sum of two sinusoids at  $f \in \{0.01, 0.03\}$  cycles per sample with amplitudes  $\{1.5, 0.7\}$ , oscillating about a physical population mean of 2.5 (well within the reachable range of  $[0.24, 5.4]$ ). Three controllers were implemented against this model: Linear-Quadratic-Gaussian (LQG), LQG with integral action (LQI), and Model Predictive Control (MPC).

All three use a Kalman filter for online state estimation. The LQR tracking cost is a per-neuron cost  $C^\top C$  ( $\lambda_{\text{var}} = 1$ ), driving all 16 output channels simultaneously. LQI adds an integrator on the centred output error to reject steady-state bias; the integrator weight  $q_{\text{int}}$  is kept small so that sinusoidal tracking error is not accumulated. MPC solves a condensed receding-horizon quadratic programme ( $H = 30$  steps, covering one full period of  $f = 0.03$ ) with explicit input box constraints, warm-started between successive steps. Controller hyperparameters were selected via a log-uniform random search (8 trials per controller) scored on RMS tracking error.



**Figure 5:** Per-neuron frequency selectivity for inputs  $u_1$  (top row) and  $u_2$  (bottom row). Each panel shows  $|H(e^{j\omega})|$  for all  $n_y = 16$  neurons (vertical axis) across  $\omega \in [0, \pi]$  rad sample $^{-1}$  (horizontal axis), on a shared logarithmic colour scale. Leftmost column: empirical response (honest OLS estimate from a single 5000-step recording, DFT zero-padded to 512 points). Columns 2–4: analytic ERA+EM transfer function  $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$  for  $n_x \in \{2, 4, 6\}$  respectively, showing the latent dimension sweep. Rightmost column: CVA+EM at  $n_x = 4$ , for estimator comparison at the chosen order. [Source](#)



**Figure 6:** Closed-loop spectral control using the ERA+EM  $n_x = 4$  model ( $n_x = 4$ ), controller hyperparameters tuned by random search (8 trials each). The target is a sum of two sinusoids at  $f \in \{0.01, 0.03\}$  cycles per sample with amplitudes  $A \in \{1.50, 1.50\}$ , centred on a physical mean of 4. Each column shows one controller (LQG, LQI, MPC) with a per-neuron tracking cost ( $\lambda_{\text{var}} = 1$ ). Row 1: physical population mean  $\bar{y}(t)$ , uncontrolled baseline (grey dashed), and target reference (black dashed). Row 2: per-neuron observations  $y(t)$  as a neuron  $\times$  time heatmap. Row 3: applied inputs  $u(t)$ . Row 4: per-neuron amplitude spectral density (neuron  $\times$  angular frequency  $\omega$ , log scale); target frequencies shown as red dashed lines. [Source](#)